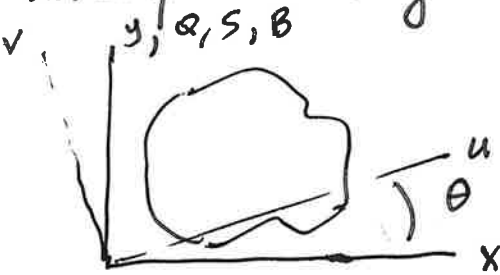
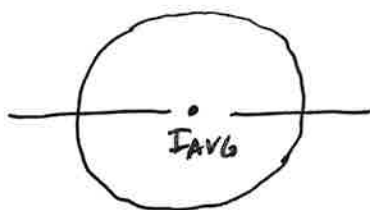


MC Exam Questions (General Principles)

Usually an original "area" or "cross section" with axes ($\perp$)
 $x-y, I-J, P-Q, \text{ Apples-Oranges, etc. } R-S, A-B$



That area or cross-section has a Mohr's Circle.

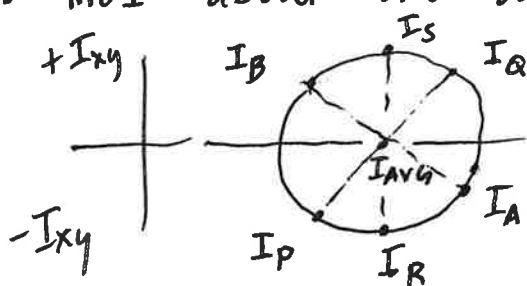


2θ on MC corresponds to θ on orig section.

Typical info given: (some combination of these)

- Center = I_{AVG}

- MOI about one axis, e.g. I_x, I_P, I_R, I_A



These can be
 $> I_{AVG}$ or $< I_{AVG}$
 or $= I_{AVG}$

- I_{xy}, I_{PQ}, I_{RS} , etc for the orig area.

- If $I_{xy} = 0$, then I_x, I_y are principals (max/min)

- If $I_{xy} \neq 0$, then I_x, I_y are not principal

- or could state that I_x, I_P etc. is max or min or principal

MC Exam Questions (cont'd)

Typical info given (cont'd)

- Could give an angle
Either θ on "cross section"
or 2θ on MC ...

from I_x, I_y, I_z etc. to I_{max} (Major Principal Axis)
or to I_{min} (Minor Principal axis)

Asked to determine:

- Radius
- I_{max} or I_{min} to "minor principal axis"
 (on Area)
- Angle to I_{min} (θ_{p2}) ($2\theta_{p2}$ on MC)
- Angle to I_{max} (θ_{p1}) ($2\theta_{p1}$ on MC)
usually smallest angle... to "major principal axis"
- Center " θ on cross section"
 (or 2θ on MC)
- I_u at some angle from I_p, I_r, I_A
 cw $\rightarrow$; ccw $\rightarrow$
- I_{uv} assoc w/ I_u at some angle
 from I_p, I_r, I_A , etc.

Example MC Problems (Extra)

①

$$\text{Center} = 20 \text{ in}^4$$

$$I_A = 12 \text{ in}^4 \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{Principal}$$

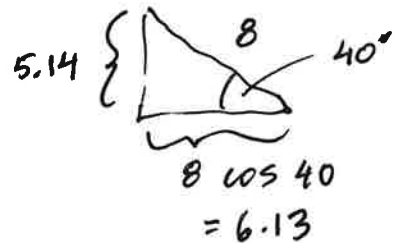
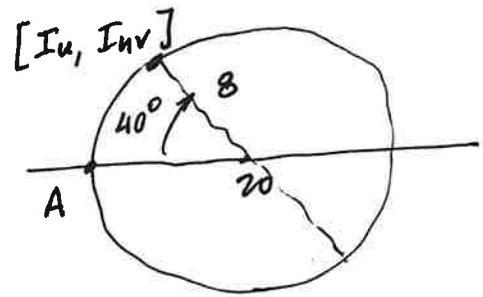
$$I_{AB} = 0$$

Find I_u 20° CW from A.

$$R = 20 - 12 = 8$$

$$I_u = I_v = 20 - 6.13 = 13.87 \text{ in}^4$$

$$I_{uv} = 5.14 \text{ in}^4$$



②

$$\text{Center} = 200 \cdot 10^6 \text{ mm}^4$$

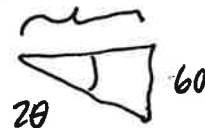
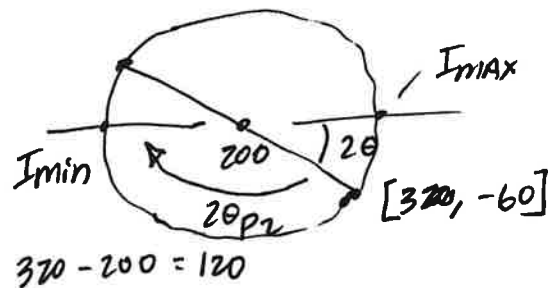
$$I_p = 320 \cdot 10^6 \text{ mm}^4$$

$$I_{pq} = -60 \cdot 10^6 \text{ mm}^4$$

Find θ to the minor principal axis:

$$2\theta_{p2} = 180 - 26.6 = 153.4^\circ$$

$$\theta_{p2} = \frac{153.4}{2} = 76.7^\circ$$



$$\tan 2\theta = 60/120$$

$$2\theta = 26.6^\circ$$

$$\theta_{p1} = \theta = 13.3^\circ$$

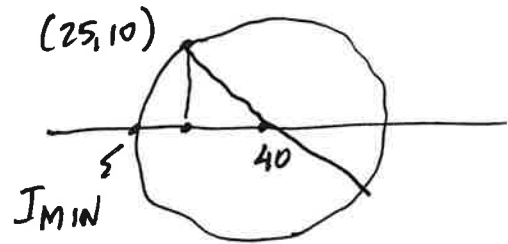
To I_{max}
so this
is θ_{p1}

$$90 - 13.3 = 76.7^\circ$$

③ Center = 40 in^4

Reference Point = $(25, 10) \text{ in}^4 = (I_P, I_{PQ})$

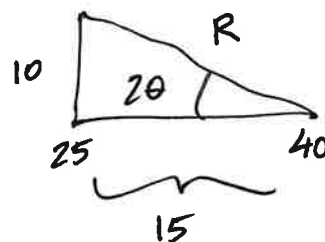
Find $I_{\min}$.
(Smallest principal moment of inertia)



$$I_{\min} = 40 - R$$

$$= 40 - 18$$

$$I_{\min} = 12 \text{ in}^4$$



$$R = \sqrt{15^2 + 10^2}$$

$$= 18,0 \text{ in}^4$$

$$\tan 2\theta = 10/15$$

$$2\theta = 33.7^\circ$$

$$\theta = 16.85^\circ$$